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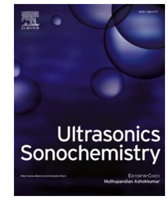

Investigations into wetting and spreading behaviors of impacting metal droplet under ultrasonic vibration control

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ABSTRACT

Ultrasonic-assisted metal droplet deposition (UAMDD) is currently considered a promising technology in droplet-based 3D printing due to its capability to change the wetting and spreading behaviors at the droplet-substrate interface. However, the involved contact dynamics during impacting droplet deposition, particularly the complex physical interaction and metallurgical reaction of induced wetting-spreading-solidification by the external energy, remain unclear to date, which hinders the quantitative prediction and regulation of the microstructures and bonding property of the UAMDD bumps. Here, the wettability of the impacting metal droplet ejected by a piezoelectric micro-jet device (PMJD) on non-wetting and wetting ultrasonic vibration substrates is studied, and the corresponding spreading diameter, contact angle, and bonding strength are also discussed. For the non-wetting substrate, the wettability of the droplet can be significantly increased due to the extrusion of the vibration substrate and the momentum transfer layer at the droplet-substrate interface. And the wettability of the droplet on a wetting substrate is increased at a lower vibration amplitude, which is driven by the momentum transfer layer and the capillary waves at the liquid–vapor interface. Moreover, the effects of the ultrasonic amplitude on the droplet spreading are studied under the resonant frequency of 18.2–18.4 kHz. Compared to deposit droplets on a static substrate, such UAMDD has 31% and 2.1% increments in the spreading diameters for the non-wetting and wetting systems, and the corresponding adhesion tangential forces are increased by 3.85 and 5.59 times.

1. Introduction

Droplet-based 3D printing, a significant subset of metal additive manufacturing (AM), promotes the development of the digital manufacturing industry with its flexibility in electronic packaging, defect repairs, microelectronic sensors, and flexible circuit preparation [1–4]. This 3D printing process is featured by low energy consumption, simple equipment, and environment-friendly [5–7]. Meanwhile, the instant solidification and the large temperature gradient [8] during the metal droplet deposition (MDD) could increase the printing efficiency. However, these printing features also induce undesirable wetting behaviors [9–10] and poor bonding mechanical properties [11–12] for the impacting MDD with large surface tension, particularly the reliability problems caused by the different wettability between the metal droplet and solid substrate with various materials and surface morphologies. These challenges are hindering the improvement and application of droplet-based 3D printing. Therefore, new techniques to enhance the wettability and bonding strength between the metal droplet and solid substrate have attracted extensive attentions [13–16].

Generally, molten metals have poor wettability on stainless, carbon, and graphite [17–19] compare to red copper and pure iron. Several studies have attempted to use various methods to improve the wettability and bonding strength between molten metals and solid surfaces. Raising the contact temperature has proven to be effective to reduce the contact angle and increase the interfacial bonding strength [20–21]. To control the dynamic behavior of the spreading metal droplet and increase the bonding strength, Zuo et al. [11] and Chao et al. [22] controlled the wettability and dynamic spreading process of the droplet by increasing the substrate temperatures to 350℃ and 650℃, respectively. However, the wetting of the droplet on the substrate was stayed at a higher temperature in a protective atmosphere, it might cause the deformation of the substrate and the remelting of the solidified droplet, which finally limited the industrial application of the droplet deposition. The more extensively utilized methods, introducing active elements such as Ti, Cr, and V into the molten metal [23–25] or adding the Ag [26], silicone [27] coatings, on the substrate are used to improve the

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wettability of liquid/ solid systems. Nevertheless, the major shortcomings are inevitable contamination and physical property changes of the molten metal. In addition, direct current control as a new type of auxiliary processing technology, named electrowetting, has been widely used in the solidification of molten metals [28–29]. Conrad et al. proposed that the microstructure and mechanical properties can be controlled by changing the recrystallization kinetics during solidification with the application of pulsed current [30–31]. However, the experimental design of applying the electric field to the molten metal is very complicated and requires specific substrate materials. Thus, a flexible, universal, and simple modification technique is required to tailor the surface wettability and enhance the bonding property.

The application of high-intensity ultrasonic to the soldering can noticeably improve the wettability of the liquid/solid system and affect the bonding properties [32–37]. Given this, the influence of surface vibration on a static droplet can be summarized as the change of the dynamic contact angle and surface energy gradient for the moving microfluid. Urai et al. measured the transient variations of the contact angle (water and ethylene glycol droplet), and an advancing contact angle decreased under the action of ultrasonic vibration [38]. Deepu et al. [39] and Manor et al. [40] presented the spreading dynamics of the static droplet on a vibration surface; it revealed that the wettability was enhanced by the momentum generation and transmission under vibration at the water-substrate interface, and the action of capillary waves at the water–vapor interface on droplet spreading was also studied. Li et. al. [41] studied the ultrasonic wetting of static Al droplet on SiC ceramic substrate. The wettability of liquid metal was improved on the ultrasonic vibration substrate at 20 kHz. Guo et al. investigated the wetting of Pb droplets on the Al surface by analyzing the contact angle using the molecular dynamic simulation, and concluded that the improvement of wettability was caused by the enhancement of diffusion and dissolution under ultrasonic vibration [42]. Yu et al. revealed that the spreading and bonding characteristics of the static Sn-Ag-Ti (-Al) solder droplets were improved with the cavitation process on an ultrasonic vibration substrate [43]. And the promotion of ultrasonic cavitation [44] on metallurgy reaction and chemical reaction was also verified. Moreover, the dynamic behaviors of the impacting droplet are significantly influenced by the ultrasonic vibration. Zhang et al. presented the control of spreading, atomization, and anti-icing of the impacting water droplet by using ultrasonic vibration [45]. In summary, numerous studies have been presented to demonstrate the adhesion between two plates, the dynamic and wetting behaviors of the non-metallic droplet or a static metal droplet on an ultrasonic substrate, however, the dynamic behaviors of an impacting droplet, particularly the metal droplet with the non-negligible solidification force, large surface tension, and enhanced metallurgical reaction on ultrasonic vibration substrate are still unclear and remain to be studied sufficiently. Therefore, the wetting and spreading dynamic behaviors of the impacting metal droplet under ultrasonic-assisted and its effect on the bonding strength of the solid surface should be described systematically.

In this study, we focus on the wetting and spreading dynamic behaviors of the impacting metal droplet with a certain velocity on ultrasonic vibration substrate with different amplitudes, and the bonding properties between the droplet and substrate are also discussed. For this purpose, there are two sets of substrate materials to analyze the wetting and spreading processes of the impacting metal droplet on non-wetting and wetting ultrasonic vibration substrates. The spreading model is proposed to analyze the ultrasonic-assisted spreading mechanism and fluctuating surface tension of the metal droplets on different wetting surfaces. A fluid-thermal-structure interaction (FTSI) simulation model is established to study the movement of the transient contact region and flow field of the impacting droplet on the ultrasonic vibration surface. Ultimately, an experimental system is designed, including the ejection of the droplet based on our previous on-demand piezoelectric micro-jet device (PMJD) [46], and the ultrasonic vibration substrate based on the self-developed longitudinal transducer (LT) [47], to validate the simulation results for the evolution of spreading diameter and contact angle.

2. Experimental system and approach

2.1. Experimental system

Fig. 1(a) shows the schematic of the established experimental system, which mainly contains three parts: the self-developed on-demand ejection device of the metal droplet with PMJD, the deposition device with LT, and the low oxygen environment. The PMJD mainly consists of a piezoelectric (PZT) stack, a flexible hinge, a transfer bar, a crucible, a base, and three cylindrical heaters. In this work, the waveform generator (RIGOL, DG4162) is used to produce a trapezoidal pulse signal formulated with a rising edge T_{rise}, dwell time T_{dwell}, and a falling edge T_{fall}. The signal is amplified via a power amplifier (Aigtek, ATA4052) and then trigger the piezoelectric stack to generate the periodic displacement. The flexible hinge and transfer bar which connect the piezoelectric stack and the diaphragm (65 Mn) are used to amplify and transfer the displacement; the diaphragm is pressed on the crucible with a conical cavity and an orifice diameter of 300 μm. The crucible is custom-made of stainless steel; the temperature of the crucible is kept and controlled with a K-type thermocouple (KAIPUSEN, TTK30SLE) and a temperature controller (SHIMAX, MAC50A). The micro metal droplet is generated from the nozzle with the diaphragm deformation and cavity volume change. The setup of the droplet deposition device mainly includes two brackets, a flange, and a LT. The flange is supported by bracket-I and bracket-II; the LT is clamped on the flange and the clamping position is located at the node of the LT's first-order longitudinal mode; the substrate is threaded on the free end of the LT. A sinusoidal voltage signal is used to trigger the LT via an ultrasonic power (QD-8D).

According to the wetting characteristics between the molten metal and the substrate, the materials of the substrate can be classified into non-wetting and wetting materials. The sketches of two different wetting behaviors are shown in Fig. 1(b). It is well known that most molten metals do not wet the stainless steel, graphite, and glasses; but most molten metals wet the red copper, H62 brass, and pure iron [17–19]. In this work, to study the effects of the ultrasonic vibration on different wetting behaviors, the non-wetting and wetting substrates are made of glass and T2 copper with Ra = 0.8 μm to investigate the changes of surface wettability with ultrasonic vibration.

To facilitate the process of the experimental operation, an acrylic plastic box with the partial opening is built to maintain the hypoxic environment. The argon gas (99.99% purity) is continuously transported from the bottom of the box to the inside during the experiment.

2.2. Experimental approaches

In this work, pure Sn is chosen for experimental metal material, and the physical properties of the Sn droplet are listed in Table 1. The initial temperature of the droplet is maintained at 548 K. Here, to avoid the effects of the residual molten Sn and the machining errors of the nozzle on the vertical drop and impact processes, the deposition distance is fixed at 5 mm. From our previous work [46], the droplet size, initial velocity, and the corresponding impacting velocity can be changed by modulating the voltage amplitude and the trapezoidal pulse signal. Thereby, the voltage amplitude and the rising time are set to 20 V and 1.1 ms, respectively. The initial velocity is 1.75 m/s. And a droplet generation frequency of 2 Hz is used.

The vibration characteristics and the dynamic displacement response of the substrate are measured by a laser Doppler vibrometer (PSV-400-M2) and a laser sensor (LK-H020) with the resolution of 5 nm and a range of 3 mm. As shown in Fig. 2(a), the copper substrate is threaded and connected at the end of the LT, and the surface of the glass and copper substrates are selected as the test area, as seen in Fig. 3, the

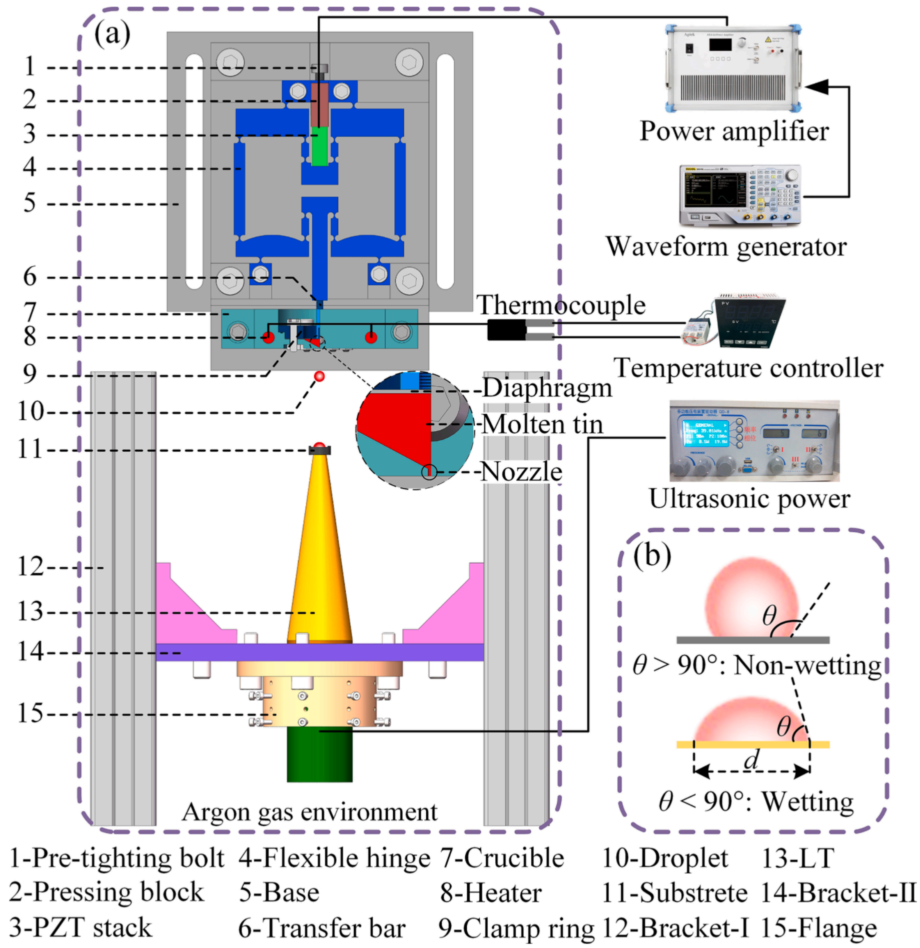

Fig. 1. Schematic diagram of the UAMDD experimental system. (a) Experimental setup based on the PMJD and LT, (b) the detailed sketch of the shape of the metal droplet with two different wetting behaviors on different substrates.

Table 1
Physical properties of Sn droplet.

Symbols	Parameters	Values	Units
ρ ρ (kg • m^{-3})	Density	7000 (T = 673 K)	kg·m^{-3}
μ μ (Pa • s)	Dynamic viscosity	1 × 10^{-3} (T = 673 K)	Pa·s
T_m	Melting point	505.11	K
σ (γ_{LV})	Surface tension	0.57 (T = 673 K)	N·m^{-1}

roughness values of copper and glass substrate are 0.003 μm and 0.079 μm, which is much smaller than the critical value of 1 μm [48]; therefore, the effect of substrate roughness on the droplet deposition can be neglected in this work. Firstly, the vibration velocity response spectrum of the ultrasonic vibration substrate is extracted and plotted in Fig. 2(b). The results indicate that the resonance frequency of the first-order longitudinal vibration is about 18.3 kHz. Then the relationships between the vibration amplitude and frequency of the substrate for four different driving voltages are described in Fig. 2(c). The tested results indicate that the vibration amplitude increases until the vibration frequency of 18.2 kHz and 18.4 kHz for the voltages of 100, 150, and 20, 50 V, then the amplitudes decrease. And the maximum vibration amplitude of the substrate is 2.38, 2.62, 8.82, and 10.94 μm at 20, 50, 100, and 150 V.

The impacting dynamic behaviors of the metal droplet on ultrasonic vibration substrate are recorded by an imaging system, which is built with an industrial camera (MERCURY, 860U3M) and a stroboscopic light (SH, B50 × 50 W) connected to a light dual channel controller (SH, STS24). An optical stereo microscope with self-built darkfield microscopy is used to get the solidification parameters (spreading diameter d, contact angle θ_s) of the droplet.

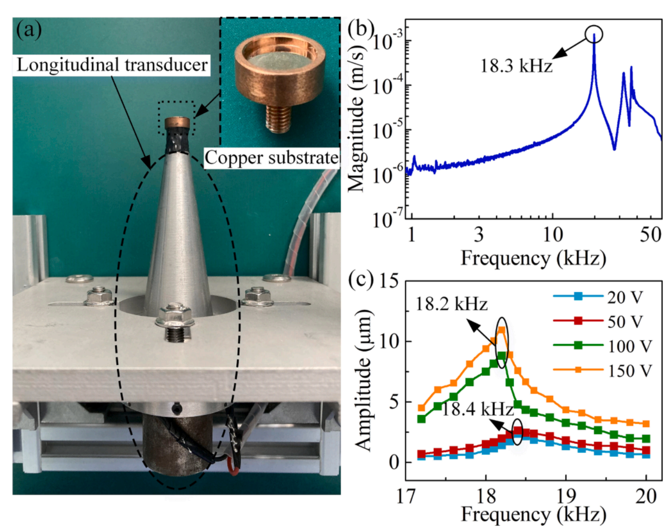

Fig. 2. Vibration test of the substrate. (a) Prototype of the ultrasonic-assisted metal droplet deposition system, (b) Velocity response spectrum of the longitudinal transducer with the substrate, (c) Relationship between vibration amplitude and frequency under different voltages.

In addition to the image processing method, the adhesion tangential

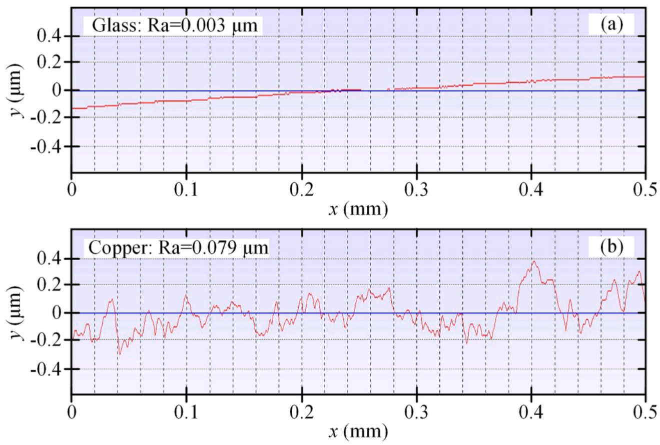

Fig. 3. Examples of line roughness curves on glass (a) and copper substrates (b).

force F_a is measured to quantify the adhesion force between the solidified Sn droplet and substrate, which is used to study the wettability and bonding strength of the droplet based on ultrasonic-assisted metal droplet deposition (UAMDD). As shown in Fig. 4, a simple adhesion tangential force measurement system is built with a 3D motion platform, a pressure sensor (GJBHX-III), a scraper, a transmitter (GJ-4057A), and a digital input module (NI-9215). First, the scraper, fixed on the pressure sensor, is adjusted right below the Sn droplet on a vibration substrate. Then the measurement program based on the LabVIEW is open while the scraper moved up towards the droplet. The data acquisition process is finished when the droplet is separated from the substrate. During the process, the adhesion tangential force is measured and amplified by the pressure sensor and the transmitter, and then the voltage signal is analyzed and recorded by the digital input module. The final data storage and waveform display are finished by the LabVIEW program on a computer.

3. Mechanism of wetting and spreading of impacting metal droplet

3.1. Wetting mechanism of impacting metal droplet

Wettability is one of the important physical and chemical properties of solid materials, which is usually measured with the stable contact angle for the droplet on solid surface [49]. At the macro scale, the contact angle is mainly measured by the interface tensions of the liquid–vapor, solid–vapor, and solid–liquid [50]. As shown in Fig. 5(b), the Young's equation, as the most fundamental theory in the field of wetting, is used to describe the balance of the above three tensions. The classical Young's equation based on the interface tensions parallel to the solid–liquid interface is formulated as:

$$\gamma_{LV}\cos\theta_{\infty} = \gamma_{SV} - \gamma_{SL} \quad (1)$$

where γ_{LV}, γ_{SV}, and γ_{SL} are liquid–vapor, solid–vapor, and solid–liquid interface tensions. θ_{∞} is the equilibrium contact angle.

In general, when a droplet impacts a solid surface, it has been proved that the spreading diameter, droplet height, and dynamic contact angle are the processes of damping oscillation with time [8]. As seen in Fig. 5 (a) and (b), the impacting droplet is subject to gravity **G**, inertial force $\boldsymbol{F}_I$, surface tension $\boldsymbol{F}_\sigma$, and viscous force $\boldsymbol{F}_\mu$ during oscillation. Therefore, violent oscillation always occurs in the impacting droplet without solidification. The three-phase contact line (TPCL) will be spreading and receding obviously. However, the solidification is instantaneous at the bottom layer of the metal droplet while the droplet begins spreading on the substrate [8]. In this way, the TPCL will be receding slightly. Thus, the TPCL of the impacting droplet is unbalanced for the spreading and

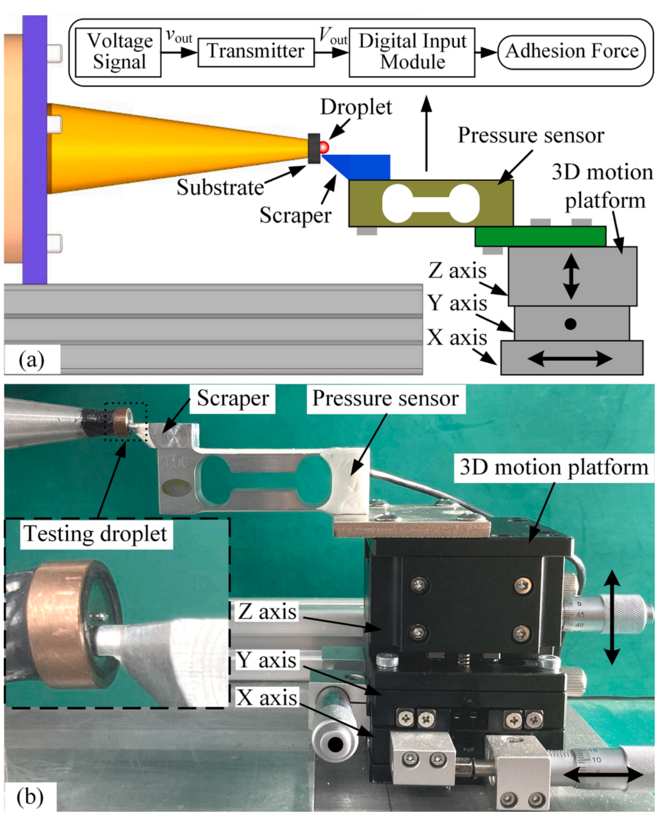

Fig. 4. Shear stress measurement system. Schematic diaphragm (a) and prototype (b) of the measurement system.

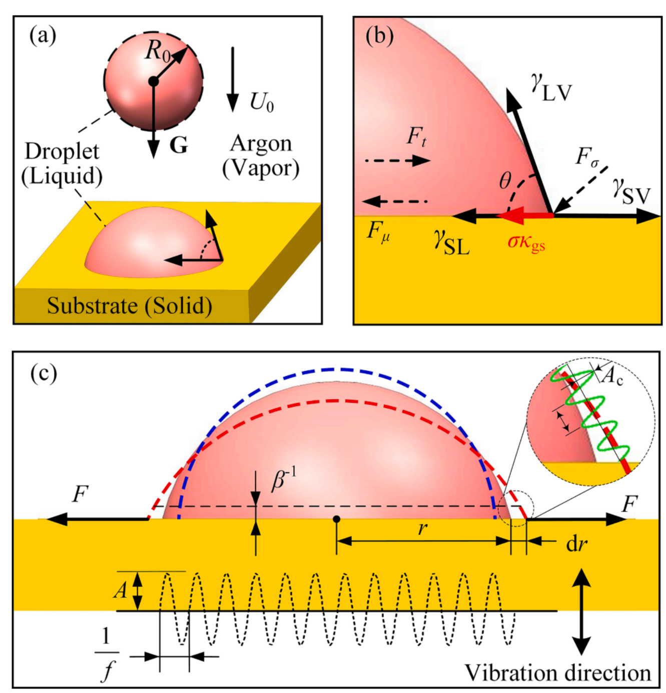

Fig. 5. Schematic of the droplet model. (a) The impacting metal droplet on a horizontal substrate, (b) Representation of Young's equation, (c) Diagram for the spreading metal droplet on ultrasonic vibration substrate with the zone change close to the TPCL (the blue and red dotted lines) and capillary waves (the green waveform). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

oscillation on the substrate. And the dynamic contact angle cannot be solved by the classical Young's equation.

In this regard, to analyze the changes in the dynamic contact angle of the droplet, the line tension is introduced based on the classical Young's equation. The interfacial tensions close to the TPCL will be balanced again under the action of linear tension, and thereby, based on the balance condition of forces, the first-order modified Young's equation is formulated as:

$$\gamma_{LV}\cos\theta + \gamma_{SL} + \sigma\kappa_{gs} - \gamma_{SV} = 0 \tag{2}$$

which θ and κ_{gs} are the dynamic contact angle and geodesic curvature of the TPCL. σ is the term used as apparent line tension [49]. The line tension is numerically equal to the free energy per unit length of the TPCL, and its direction points to the center of the circle, which tends to ensure the shortest length of the TPCL. In this case, we assume that the spreading processes of the impacting droplet are axisymmetric, thereby, the geodesic curvature is given as:

$$\kappa_{gs} = \frac{1}{r} \tag{3}$$

where r is the contact radius. Then the dynamic contact angle can be described as:

$$\cos\theta = \frac{\gamma_{SV} - \gamma_{SL}}{\gamma_{LV}} - \frac{\sigma\kappa_{gs}}{\gamma_{LV}} = \cos\theta_\infty - \frac{\sigma}{\gamma_{LV}}\frac{1}{r} \tag{4}$$

Eq. (4) shows that the contact angle will vary with the radius of the TPCL. For a given solid–liquid system, $\cos\theta_\infty$ is a constant, and $\cos\theta$ should be a linear function of $1/r$.

3.2. Two spreading behaviors under ultrasonic vibration

When the ultrasonic vibration is applied to the solid surface, a viscous momentum transfer layer, induced by the ultrasonic shear waves [51], is created at the solid–liquid interface. The momentum transfer layer possesses the characteristics of the Schlichting boundary layer. It interacts with the droplet surface near the TPCL, and an additional tension is induced, which can push the TPCL outward [52]. Similar to the action mechanism of the impact force of the droplet with initial velocity on the substrate, the dynamic TPCL under the ultrasonic action will also reach a new equilibrium position. As shown in Fig. 5 (c), the new correction of Young's equation is formulated as:

$$\gamma_{LV}\cos\theta + \gamma_{SL} - \gamma_{SV} - F = 0 \tag{5}$$

where F is the additional acoustic tension induced by ultrasonic vibration. The interfacial tension created by the viscous momentum transfer layer is located in the submicron region, and the thickness of this region can be expressed as:

$$\begin{cases} \beta^{-1} = \sqrt{2\mu/\rho\omega} \\ \omega = 2\pi f \end{cases} \tag{6}$$

where μ and ρ are the viscous and density of the droplet. ω and f are the angular frequency and vibration frequency of the substrate. In this work, f is 19746 Hz, μ and ρ are defaulted to 1.3×10^{-3} Pa•s and 7280 kg•m^{-3}, respectively. Based on Eq. (6), the value of β^{-1} for molten Sn is about 1.697 μm in a manner analogous to Schlichting streaming.

The radical acoustic tension per unit length along the contact line has been proposed by Manor *et al.* [42], which is calculated as:

$$\begin{cases} F \approx \dfrac{\rho}{32\beta^{-1}}(U_c r)^2\cos^2\theta \\ U_c = \omega\xi_0 \end{cases} \tag{7}$$

where U_c is the streaming velocity at the bottom layer of the metal droplet. ξ_0 is the surface vibration half-amplitude. Substituting Eq. (7) into Eq. (5) yields the following equation:

$$\gamma_{LV}\cos\theta + \gamma_{SL} - \gamma_{SV} - \frac{\rho}{32\beta^{-1}}(U_c r)^2\cos^2\theta = 0 \tag{8}$$

The singular solution of Eq. (8) is $\theta = 90^\circ$, which means that the TPCL of the droplet will not be affected by the additional acoustic tension when the equilibrium contact angle θ_∞=90°. However, for the dynamic spreading process of the impacting metal droplet on ultrasonic vibration substrate, based on the balance condition of forces in Eqs. (2) and (8) near the TPCL, a second-order correction of Young's equation is formulated as:

$$\gamma_{LV}\cos\theta + \gamma_{SL} + \sigma\kappa_{gs} - \gamma_{SV} - \frac{\rho}{32\beta^{-1}}(U_c r)^2\cos^2\theta = 0 \tag{9}$$

Substituting Eqs. (1), (3), and (6) into Eq. (9), then the dynamic contact angle is described with the following equation:

$$\cos\theta - \cos\theta_\infty = We\frac{r}{\sqrt{\mu/\rho\pi f}}\frac{\cos^2\theta}{64} - \frac{\sigma}{\gamma_{LV}}\frac{1}{r} \tag{10}$$

where We is used to present Weber number which defines the ratio of inertia force to surface tension, $We = \rho U_0^2 r/\gamma_{LV}$. For the impacting and spreading processes of the droplet in this work, We is concerned with the mechanical power of the ultrasonic vibration which transfers into the liquid through the wetting region at the bottom of the droplet.

As shown in Fig. 6, it is assumed that ΔE_{dim}=(cosθ-cosθ_∞) which is described by the dynamic contact angle θ. The dashed lines in Fig. 6 are used to describe the sum of the surface tensions $\gamma_{LV}\cos\theta + \gamma_{SL}$-$\gamma_{SV}$ on the left of Eq. (10). The colored areas are used to describe the right items of Eq. (10), which are defined by the energy injected by ultrasonic vibration and line tension of the impacting droplet. The boundaries of these areas are controlled by the impacting velocity (0.56 ~ 2.13 m/s, corresponding initial velocity 0.51 ~ 2.12 m/s [46]) and the contact radius under non-wetting and wetting conditions. For the non-wetting system, as shown in Fig. 6(a), there is a narrow area near the contact angle $\theta = 90^\circ$ (the green shadow), and the left terms in Eq. (10) are larger than the right for $90^\circ < \theta < 105^\circ$. Thereby, the spreading of the droplet can be driven by the ultrasonic vibration when for $\theta > 105^\circ$. For the wetting system, the right terms in Eq. (10) are always larger than the left. Therefore, the spreading processes of the droplet are affected by the ultrasonic vibration for the wetting system. In addition, it can be found that the influence of surface tension on droplet spreading is probable small when We is relatively large for the non-wetting system and wetting system. And values of 7.47 and 15.79 (Weber number, orange solid lines in Fig. 6) are chosen for non-wetting system and wetting system in this work, respectively.

As shown in Fig. 5(c), with the injection of ultrasonic vibration energy, the liquid–vapor interface and TPCL of the spreading droplet also have small vibrations. In addition, the capillary waves on the surface of the droplet are created along with these vibrations. Although the surface tension, as the inherent property of the liquid, cannot be changed, the superficial area at liquid–vapor surface of the droplet increases significantly along with the capillary waves [49]. The interfacial roughness caused by vibration will lead to a change of the free energy at TPCL. Then, the change of the free energy per unit length can be described as:

$$C_W\gamma_{LV}(\cos\theta_s - \cos\theta_\infty) = \mathrm{d}G/\mathrm{d}r \tag{11}$$

where C_W is used to define the ratio of the waveform length A_L of capillary wave to its wavelength. In this work, θ_s is the contact angle of the solidification Sn droplet on ultrasonic vibration substrate. The waveform length A_L during a single cycle is approximately expressed as:

$$\begin{cases} A_L = 4\sqrt{A_c^2 + (\lambda/4)^2} \\ \lambda = 2\pi/k \\ \omega_c = \gamma_{LV}k^3/\rho \end{cases} \tag{12}$$

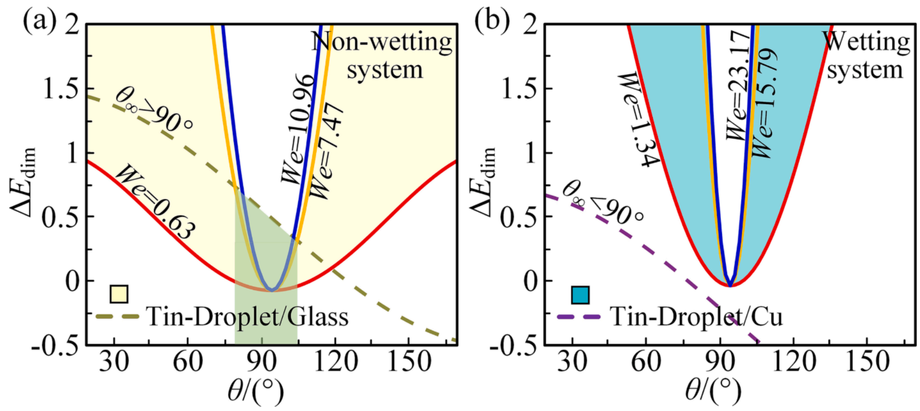

Fig. 6. Graphical solution of Eq. (10) under different impacting velocity (defined with We) and θ. (a) The non-wetting system for the Sn droplet and glass substrate, (b) The wetting system for the Sn droplet and copper substrate.

where A and λ are the amplitude and wavelength of the capillary waves. k is the wave number which is related to the angular frequency ω_c of the capillary waves. In this work, whether a non-wetting or wetting system, it is assumed that the left and right sides of the spreading droplet on a horizontal vibration substrate are symmetrical (Fig. 5(c)). It can also be defined as the pinning behavior at the TPCL. In this case, the free energy at two sides of the droplet is almost equal, $[dG/dr]_L \approx [dG/dr]_R$. Therefore, the ultrasonic vibration energy injected into the droplet will drive the TPCL of the droplet to move symmetrically along the substrate surface.

4. Results and discussions

4.1. Spreading characteristics of metal droplet on an ultrasonic substrate

4.1.1. Multi-physics numerical model validation

A direct multi-physics FTSI numerical model has been established in our previous work [46]. In this work, the FTSI model is revised to simulate the spreading dynamics of the metal droplet on the FTSI surface with ultrasonic vibration in the commercial software ANSYS Fluent. The molten metal is a typical Newtonian fluid, and the flow is assumed to be laminar and incompressible, which is governed by the conservations of mass, momentum, and energy. For the UAMDD process, a non-inertial reference frame is laid on the vibration surface. The inertial force induced by ultrasonic vibration is integrated into the momentum equation, seen in Eq. (13).

$$\frac{\partial}{\partial t}(\rho \boldsymbol{v}_f) + (\boldsymbol{v}_f \cdot \nabla)\rho \boldsymbol{v}_f = \nabla \cdot (\mu \nabla \boldsymbol{v}_f) - \nabla p + \boldsymbol{f}_B + S_D + S_u \tag{13}$$

where $\mathbf{v}_f$ is the fluid velocity in flow field, p denotes the pressure, $\mathbf{f}_B$ is the buoyant force, and S_D is used to define the Darcy force. S_u is the inertial force induced by the ultrasonic vibration surface, which is defined by the following equation:

$$S_u = -\rho f 2A\sin(ft) \tag{14}$$

In the simulation, the FTSI surface is used to define the vibration substrate in experimental setup. The ultrasonic vibration frequency f (resonance frequency) and amplitude A (defined in Fig. 5) remain

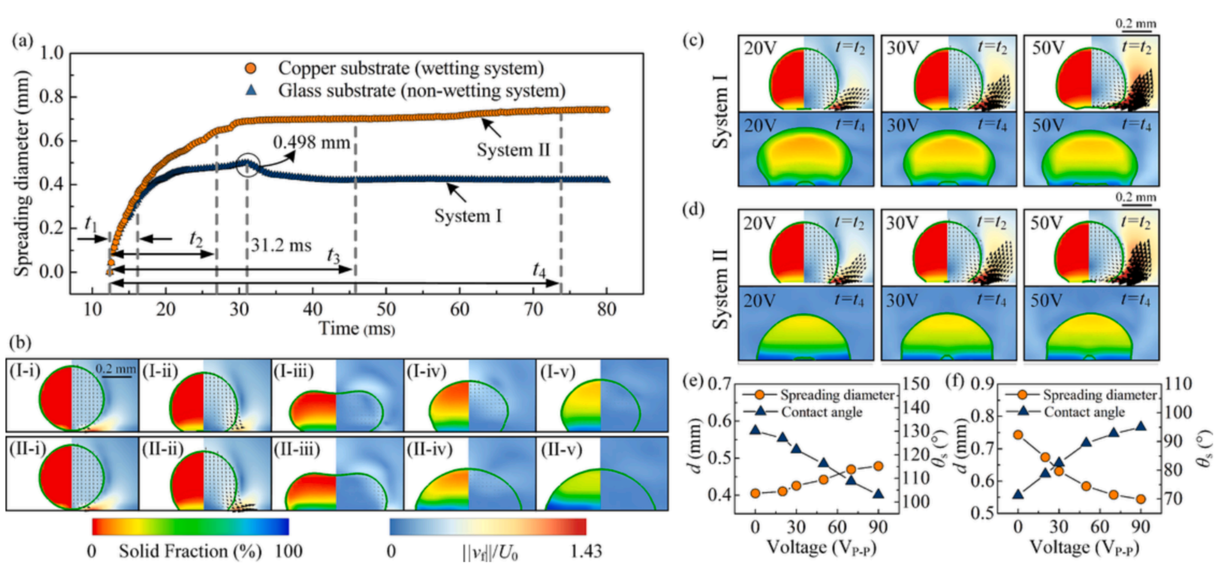

Fig. 7. Simulation analysis of the metal droplet spreading dynamic with and without ultrasonic vibration. (a) Spreading diameter d(t) on a static substrate as a function of time: I is used to define the non-wetting system, and II is used to define the wetting system. (b) Simulation results for a drop impact dynamic for $D_0 = 520$ μm and $U_0 = 1.75$ m/s: t = (i) 12.4 ms (touchdown), (ii) 16 ms, (iii) 27.2 ms, (iv) 46 ms, (v) 74 ms. The left part of each simulation snapshot displays the dimensionless local solid fraction of the molten metal and the right part the velocity field magnitude of the liquid metal normalized with the impact velocity. The black velocity vectors are extracted and drawn in the center of the mass reference frame of the droplet and the right side of the contact region to clearly demonstrate the flow of non-solidified metal and ambient flow. (c) and (d) Numerical snapshots for the spreading droplet on an ultrasonic substrate. (e) and (f) The dynamic spreading diameter and contact angle versus the voltages applied on the ultrasonic substrate.

constant which is determined by transient analysis of the LT.

4.1.2. Spreading and flow characteristics

The wettability of the metal droplet on glass and copper substrate is defined by the inherent interface contact angle between materials. Fig. 7 (a) quantifies the spreading diameter $d(t)$ (the maximum contact width of the droplet at time t) of the metal droplet on the non-wetting glass substrate (defined as system I) and wetting copper substrate (defined as system II) without ultrasonic vibration. As the droplets touch the static substrate (Fig. 1(b)(I-i) and (II-i)), note the quantitative agreement between the systems I and II during the first 3.6 ms, the spreading diameters increase obviously. However, contrary to the velocity field in Fig. 1(b)(I-ii) at $t_1 = 16$ ms on the non-wetting system, for the system II, the velocity vector shows sharper. It indicates that the tendency of the spreading metal droplet to both sides is stronger for the wetting system. Subsequently, during the further spreading stage, compared with the steadily increasing of the system I, the spreading diameter of the system II increases obviously owing to the given small interface contact angle and the traveling capillary waves for 16 ms (t_1) < t < 27.2 ms (t_2). Meanwhile, as shown in Fig. 1(b)(I-iii) and (II-iii), the first stroke of the droplet spreading process is finished at $t_2 = 27.2$ ms for system I and system II which have the spreading diameters of 478 μm and 648 μm. During the third stage (27.2 ms (t_2) < t < 46 ms (t_3)), the droplets start to go up for both systems. In this stage, the droplet for system I reaches a maximum spreading diameter $d_{\max} = 498$ μm at time $t = 31.2$ ms, which is different from the continuous but slow increase of the spreading diameter from that in system II. Generally, the spreading diameter cannot retract obviously due to the instantaneous solidification when the droplet touches the substrate. For system I, the poor wettability prevents the spreading of the droplet itself. The spreading diameter will retract slightly during the receding stage beginning at $t = 31.2$ ms. However, for system II, better wettability promotes the adhesion between the droplet and substrate. Therefore, the retraction of the spreading diameter for system II could be neglected. Finally, the values of the spreading diameter of both systems tend to be stable, however, different from the change of spreading diameter of system I which increases slowly and tends to be stable (405 μm), the spreading diameter of system II decreases and tends to be stable (743 μm) for 46 ms (t_3) < t < 74 ms (t_4). Generally, the transient solidifying reaction is happening when the droplet contacts the substrate at normal temperature; moreover, Sn and copper can adhere to each other due to the metallurgy reaction between the droplet and substrate. Therefore, the droplet is rarely retracting on a cooper substrate (system II). However, the bonding strength between the metal droplet and the glass substrate is weak without the chemical reaction, and the retraction process is relatively obvious. In this regard, it can be found that the spreading diameter and contact angle of the droplets for both systems change slightly after the first stroke.

Next, the specific velocity field magnitude combined with the velocity vector (black arrow, time $t_2 = 27.2$ ms in Fig. 7(c) and (d)) which are driven by ultrasonic vibration from the perspective of the flow field, which is used to verify the driving mechanism of ultrasonic vibration surface on the spreading droplet introduced in the theoretical model. As seen from Fig. 7(c) and (d), whether the non-wetting system or the wetting system, the overall dimensionless velocity field $||v_f||/U_0$ and velocity vector on both sides of the droplet are enhanced with increasing voltage at t_2. Meanwhile, it can be found that the intensity of the velocity vector field along the vertical direction of the vibration surface increases remarkably with increasing voltage. The regular direction change of the flow field near the surface is caused by the change of the near-field sound pressure induced by the ultrasonic vibration surface (The spreading and sputtering ratio of the droplet in experiments are elaborated in detail). Specifically, as shown in Fig. 7(c), the diameter of the contact region for the solidified droplet in system I increases with the increase of the applied voltage at t_4 (Time node t_4 in Fig. 7(c) and (d) is assumed to be initial solidification shape of the metal droplet on ultrasonic vibration substrate). Meanwhile, Fig. 7(e) quantifies the influence of the voltage on the spreading diameter and contact angle of the solidified metal droplet. It can be seen that the UAMDD method can enhance the wettability of the metal droplet in a non-wetting system, and the spreading diameter increases with an increasing voltage, but the morphology of the droplet tends to flatten. Here, for the non-wetting system in Fig. 7(c), the combined action between the extrusion of the vibrating substrate and the instantaneous solidification of the droplet on substrate is the main reason to increase the spreading diameter; the flattening is more obvious with the increase in voltage. The oscillation and surface tension of the upper unsolidified droplet can maintain the original appearance of the droplet and decrease the contact angle with the increase of the vibration amplitude of the substrate. In addition, the capillary wave at the liquid/vapor interface (described in Fig. 5(c)) also can be a factor of driving force, which can lead to the change of the free energy dG/dr symmetrically at TPCL. For the wetting system in Fig. 7 (d), the ultrasonic waves create a momentum transfer layer at the solid/liquid interface, which is described with the velocity vector on bottom layer of the droplet (shown at t_2), and it provides a strong driving force on the TPCL. The driving force increases with the increase of the vibration amplitude. However, the UAMDD method can suppress the wettability of the droplet in wetting system (system II, shown in Fig. 7(d) (t_4) and Fig. 7(f)) in a way. The spreading diameter does not increase obviously in simulation, the main reason is caused by the ultrasonic levitation with the vibration amplitude A which has a suspending effect for a droplet on ultrasonic vibration surface in the wetting system, and the deformed contact region will retract instantly under the large surface tension of the liquid metal. And the effect of suspension levitation becomes more pronounced as the amplitude of ultrasound vibration increases.

4.2. Microstructure characterization of solidified droplet in UAMDD

4.2.1. Test of spreading diameter on ultrasonic vibration substrate

To experimentally verify the influence of the vibration substrate on the spreading characteristic for non-wetting system and wetting system, the microstructure morphology of the fully solidified Sn droplet in UAMDD is analyzed. The ranges of voltage and corresponding working frequency applied to the LT are 10 ~ 130 V and 18.2 ~ 18.4 kHz, respectively. The testing data points for each set of parameters on LT are collected by combining 15 times of measurements.

Fig. 8 shows the spreading behaviors of the droplet in non-wetting system (System I). For the adhesive droplet, as shown in Fig. 8(a), the spreading diameter d increases with an increasing voltage. The results indicate that the ultrasonic effect on the change of spreading diameter $\Delta d = 110$ μm. Three examples of Sn bumps in UAMDD at $V_{P\text{-}P} = 30$, 70, and 120 V are also shown in Fig. 8(a), and the corresponding spreading diameters are 360, 410, and 460 μm. Based on the simulation results, as seen in Fig. 2(c), the resonant amplitude of the substrate increases with an increasing voltage. Under the action of the large amplitude and working frequency, the Sn droplet is squeezed strongly and finally results in the "flattening" (Fig. 7(c) and three examples in Fig. 8(a)); meanwhile, the liquid Sn along the contact region is forced to squeeze to both sides of the droplet. Thus, the spreading diameter of the solidified droplet increases with an increasing voltage. As seen in Fig. 8(c), some droplets cannot adhere to the substrate under the action of ultrasonic vibration starting at 80 V. And the sputtering ratio increases with an increasing voltage. In simulation, the normal spreading and solidification process are no longer exhibited when the voltage exceeds 80 V. Fig. 8(b) shows the velocity magnitude surrounding the droplet on the vibration surface in different directions at two adjacent time nodes (t_2-0.02 ms and t_2). First, the velocity field $||v_f||/U_0$ increases greatly when $V_{P\text{-}P} = 100$ V, and the maximum flow velocity $v_f = 4.57$ m/s which is far greater than $v_f = 1.02$ m/s at $V_{P\text{-}P} = 50$ V. Secondly, the uneven velocity vector along the contact region (left side of the droplet in Fig. 8(b)) due to the uneven up and down of the droplet during vibration, especially

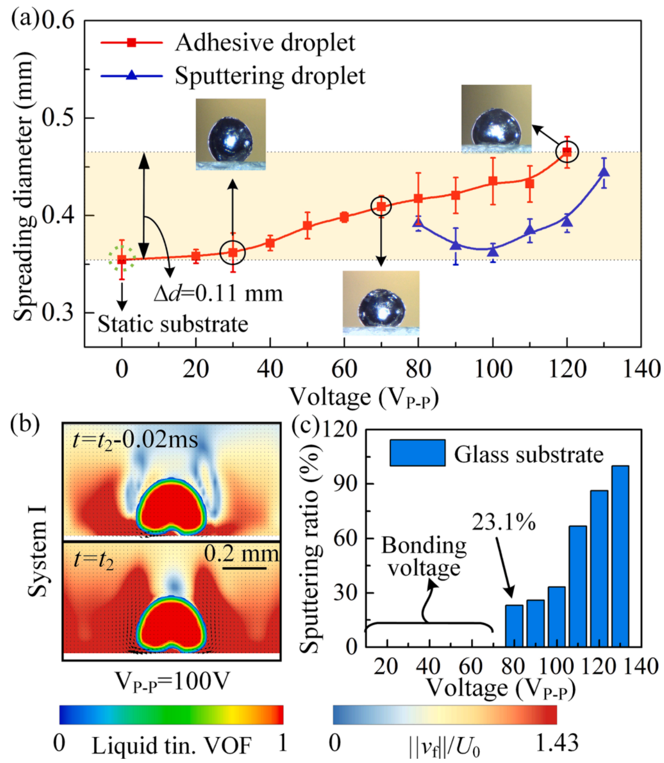

Fig. 8. Spreading behaviors of the droplet in a non-wetting system (System I). (a) Spreading diameter as a function of voltage. (b) Sputtering mechanism of the droplet at $V_{P\text{-}P}$ = 100 V. (c) Histograms of the sputtering ratio of droplets with voltages.

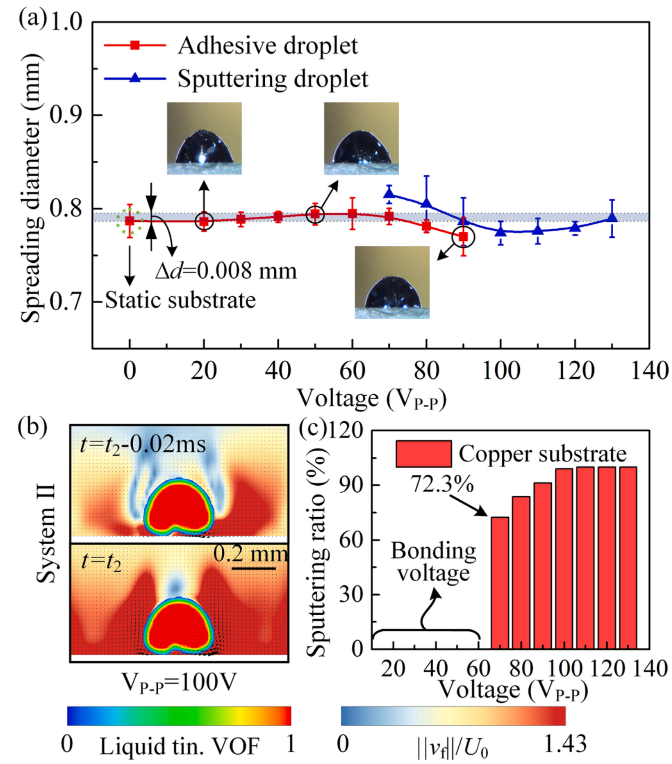

Fig. 9. Spreading behaviors of the droplet in a wetting system (System II). (a) Spreading diameter as a function of voltage. (b) Sputtering mechanism of the droplet at $V_{P\text{-}P}$ = 100 V. (c) Histograms of the sputtering ratio of droplets with voltages.

the substrate roughness is not consistent in experiment, also increases the sputtering ratio.

For the wetting system (System II), as seen in Fig. 9(a), the spreading diameter increases firstly at the voltage ranges of 0 ~ 50 V and then decreases when $V_{P\text{-}P}$ > 50 V. The results indicate that the ultrasonic effect on the change of spreading diameter Δd = 8 μm. Three examples of Sn bumps in UAMDD at $V_{P\text{-}P}$ = 20, 50, and 90 V are also shown in Fig. 9(a), and the corresponding spreading diameters are 786, 794, and 769 μm. It can be seen that the ultrasonic vibration with a small amplitude (2.38 ~ 2.62 μm) can improve the wettability of the metal droplet in the wetting system. The possible factors that affect the increase of spreading diameter mainly include the momentum generation and transmission under vibration at the droplet-substrate interface and the capillary waves at droplet-air interface are induced by ultrasonic vibration, which increases the wettability and spreading diameter of the droplet.

However, compared with the non-wetting system, Fig. 9(a) and (c) shows that the sputtering voltage decreases to 70 V, and the starting ratio for the sputtering droplets increases to 72.3%. The main reasons are as follows: First, the ultrasonic vibration substrate with the large amplitude is equivalent to adding a layer of "extra invisible roughness", and its value is approximate to the vibration amplitude. The droplet is forced to suspend under the action of the "extra invisible roughness". Therefore, the contact region between the droplet and wetting surface decreases, and the wettability decreases without the capillary force. In this regard, the spreading diameter of the suspended droplet is forced to decrease under the action of the surface tension (minimize surface area). In addition, the uneven velocity field on the bottom of the droplet also happens for the large displacement of the droplet in the wetting system, which is shown on right side of the droplet in Fig. 9(b); and the maximum flow velocity v_f = 4.63 m/s is bigger than the droplet in the non-wetting system.

4.2.2. Test of contact angle on ultrasonic vibration substrate

The final contact angle of the solidified Sn droplet (without or with UAMDD) is often measured with an optical stereo microscope and dark-field image. Fig. 10(a) and (b) shows the relation between the contact angle and the voltage, in which the ranges of voltage and corresponding working frequency of LT for the non-wetting system (system I) and wetting system (System II) are set as 0 ~ 130 V (18.2 ~ 18.4 kHz) and 0 ~ 90 V (18.2 ~ 18.4 kHz), respectively. For system I, as shown in Fig. 10 (a), it can be found that the contact angle decreases with an increasing voltage. Fig. 10(c) shows the snapshots of the solidified droplets under different voltages, and the average contact angle decreases from 142.4° to 106.3° for the ranges of voltage 0 ~ 130 V. Although the flattening of the droplet can affect the reduction of the contact angle, the contact angle still increases during the vibration extrusion and transient solidification process as well as the rapid upward retraction of the non-solidified Sn at the upper part of the droplet. However, as shown in Fig. 10(a), the contact angle of the sputtering droplet increases uniformly under the corresponding voltage; meanwhile, the sputtering droplets have a big contact angle but a small spreading diameter (shown in Fig. 8(a)). It can be drawn from Fig. 10(b) that the contact angle in system II decreases and then increases with the voltage.

Fig. 10(d) shows the snapshots of the solidified droplets under different voltages. And the results indicate that the ultrasonic effect on the reduction of contact angle $\Delta\theta$ = 8.8° which is smaller than 32.8° in system I. Therefore, the ultrasonic vibration can decrease the contact angle for the formation of the momentum transfer layer and capillary waves when $V_{P\text{-}P} \leq 50$ V; it should be noted that the simulation results are inconsistent with the experimental, this is mainly because the material settings of the FTSI surface simulation cannot be completely consistent with material properties of the substrate used in the experiment. However, the support of the ultrasonic vibration substrate and the suspension of the droplet also increase the contact angle in the wetting system. Among them, the influences of the ultrasonic vibration on

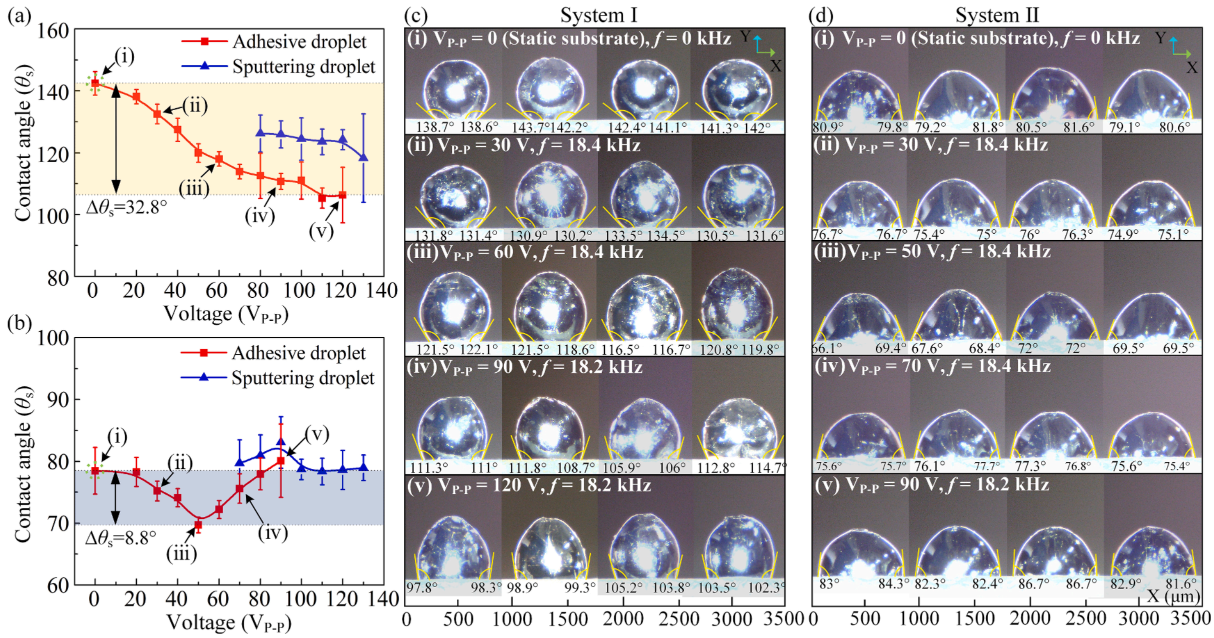

Fig. 10. Contact angle of the solidified droplet in UAMDD. Contact angle as a function of voltage in the non-wetting system (system I) (a) and wetting system (system II) (b). The snapshots of the solidified droplets under different voltages and resonant frequencies in system I (c) and system II (d).

contact angle are related to the change of spreading diameter; and the relatively low voltages (0 V < $V_{P\text{-}P}$ < 50 V) can improve the wettability of the droplet in the wetting system. Therefore, the experimental results of the spreading diameter and contact angle better verify the simulation results.

4.2.3. *Test of bonding strength on ultrasonic vibration substrate*

To verify the effect of the UAMDD method on the bonding strength between the Sn droplets, the adhesion tangential force F_t between the solidified droplet and substrate is measured in above mentioned two systems ($F_{a\text{-}s}$: the adhesion tangential force for the static substrate; $F_{a\text{-}v}$: the adhesion tangential force for the vibration substrate). Fig. 11(a) and (b) shows the changes of adhesion tangential force when measuring every single droplet deposited on a glass substrate (non-wetting surface) and a copper substrate (wetting surface). The peak value of each curve is used to represent the adhesion tangential force. As shown in Fig. 8(c) and Fig. 9(c), it can be found that the droplet cannot adhere to the substrate under the action of violent ultrasonic vibration when the applied voltage $V_{P\text{-}P}$ exceeds 120 V and 90 V for the non-wetting system and the wetting system, respectively. Therefore, the voltage test conditions for the mentioned two systems are different. Fig. 11(c) shows that the adhesion tangential force of the droplet deposited on a glass substrate ranges from 0.71 mN ($F_{t\text{-}s}$, static substrate) to 3.95 mN (A = 2.91 μm at $V_{P\text{-}P}$ = 60 V) and then decreases to 2.94 mN (A = 5.6 μm at $V_{P\text{-}P}$ = 120 V); and the adhesion tangential force of the droplet deposited on a copper substrate ranges from 1.61 mN (static substrate) to 6.16 mN (A = 2.91 μm at $V_{P\text{-}P}$ = 60 V) and then decreases to 3.69 mN (A = 4.09 μm at $V_{P\text{-}P}$ = 90 V). First, the larger adhesion tangential force of the static copper substrate is likely caused by the wettability and metallurgy reaction of Cu/Sn intermetallic compounds (Cu_6Sn_5) [29]. In addition, the results demonstrate the ultrasonic vibration can successfully enhance the bonding strength for the non-wetting and wetting systems; there are 3.85 times increase in adhesion tangential force for the non-wetting system at $V_{P\text{-}P}$ = 60 V, and 5.59 times increase for the wetting system at $V_{P\text{-}P}$ = 60 V. For the non-wetting system, the increase of adhesion tangential force is mainly caused by the increase of spreading diameter or the refinement of the contact region between the droplet and substrate (physical factors). For wetting system, the increase of adhesion tangential force is caused by the above-mentioned physical factors and

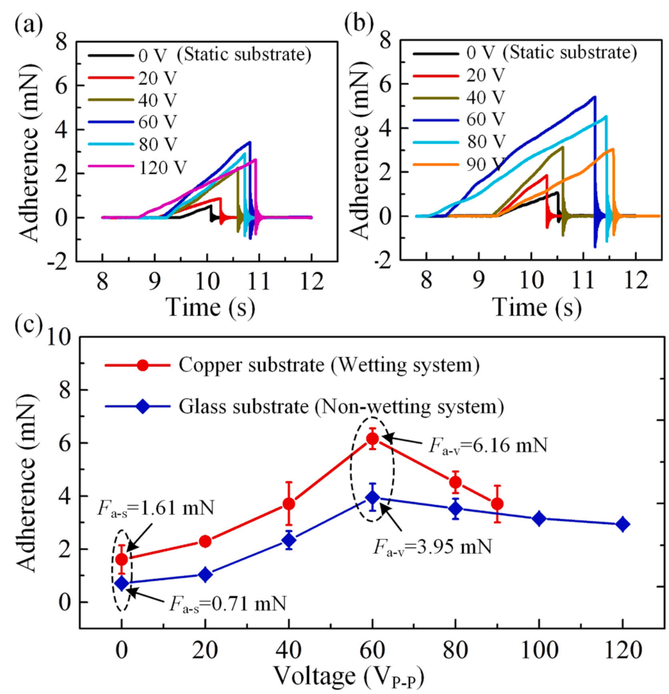

Fig. 11. Adhesion tangential force of solidified droplets in UAMDD. The deposited droplets on a glass substrate (a) and a copper substrate (b) at different voltages. (c) Adhesion tangential force as a function of voltage in the non-wetting system (glass substrate) and the wetting system (copper substrate).

the enhancement of the metallurgy reaction by ultrasonic cavitation. However, the adhesion tangential force decreases obviously owing to the decrease of the spreading diameter and the suspension of the droplet which weakens the metallurgy reactions.

The comparisons of bonding strength enhancement between the previous studies and the proposed UAMDD with a LT in this work are listed in Table 2. The adhesion increase ratio ($F_{a\text{-}v}/F_{a\text{-}s}$ in this work) is used to present the improvement times of bonding mechanical property

Table 2
Comparison of the adherence by using different assisted methods.

First author and year	Assisted methods	Material wettability	Contact form	Adhesion increase ratio
2014 [23] Koleňák	Active materials	Non-wetting	Plate-solder-plate	2.56
2019 [27] Liu	Coatings	Wetting	Static droplet-plate	2.00
2016 [11] Zuo	Temperature	Wetting	Impacting droplet-plate	2.50
2017 [20] Fu	Temperature	Non-wetting	Static droplet-plate	1.20
2005 [34] Kago	Ultrasonic pressure	Wetting	Static droplet-plate	1.50
		Wetting	Static droplet-plate	1.23
2004 [33] Xu	Ultrasonic vibration	Wetting	Plate-solder-plate	2.51
This work	Ultrasonic vibration	Non-wetting	Impacting droplet-plate	3.82
		Wetting	Impacting droplet-plate	5.59

between the solidified droplet and substrate (plate). The existing adherence enhancement methods consist of adding active materials and coatings, temperature control, and assistance with ultrasonic vibration. For example, a Sn solder alloyed with active Ti element was used by Kolenak *et al.* [23], which can increase the shear strength between Sn and SiO_2 ceramics from 6.67 MPa to 17 MPa (2.56 times increase). Liu *et al.* [27] used a supramolecular silicone coating on the glass, which can increase the shear strength of Siloxane oligomers from 10 MPa to 20 MPa (2 times increase). In addition, Zuo *et al.* [11] and Fu *et al.* [20] introduced temperature control to achieve 2.5 and 1.2 increases in tensile and bonding strength for 7075 Al/7075 Al and SnAgCu-Ti/Graphite. It is not difficult to find that the above-mentioned methods are seriously limited because of the inevitable contamination, thermal damage, and smaller increase ratio of adherence. Although the ultrasonic-assisted bonding enhancement of solder/copper and Al_2O_{3P}/Al had been proposed by Kago *et al.* [34] and Xu *et al.* [33], the contact forms mainly focus on plate-solder-plate and the static droplet-plate, and the bonding strength is improved slightly. It is worth noting that the research object of UAMDD in this study not only focus on the plate-plate or the static droplet on the substrate, and the purpose of this work is to study the spreading dynamic behaviors and bonding strength of the impacting droplet on ultrasonic substrate. In addition, the “Ultrasonic vibration” as an assisted method is precisely divided into “Ultrasonic pressure” and “Ultrasonic vibration”, “Ultrasonic pressure” means non-contact indirect assisted deposition with ultrasonic horn (Kago *et al.* [34]), “Ultrasonic vibration” means contact direct assisted deposition with ultrasonic vibration devices (Xu *et al.* [33]). The adhesion increase ratio with the assisted strategies of ultrasonic (vibration and pressure) and other three classical methods (active materials, coatings, and temperature) are contrasted firstly. It indicates that the ultrasonic can increase the adhesion during the metal droplet deposition; meanwhile, it can be found that the ultrasonic vibration (contact and direct driving method) has better adhesion enhancement than ultrasonic pressure (non-contact and indirect driving method). However, there are few researches on applying the ultrasonic vibration on deposition process of the impacting metal droplet. Finally, the proposed UAMDD, as a contact and direct driving method in this work, provides an obvious enhancement in the bonding strength for the impacting Sn droplet on a copper substrate (wetting) and a glass substrate (non-wetting). In summary, the above comparisons clearly indicate that the UAMDD method has the potential to widen its applications in 3D printing.

5. Conclusion

In this study, the ultrasonic-assisted metal droplet deposition (UAMDD) had been used to address spreading dynamics and bonding properties in droplet-based 3D printing. A spreading model was established based on revised Young’s equation to analyze the wetting and spreading behaviors, which revealed the action of ultrasonic vibration on the dynamic contact angle of the impacting metal droplet for non-wetting and wetting systems. Based on this, the fluid flow and sputtering mechanisms for the droplet on the ultrasonic vibration FTSI surface were simulated based on the VOF method; the metal droplet spreading process was restored by the velocity field magnitude and solid fraction; meanwhile, the driving mechanism of the ultrasonic vibration on the TPCL was analyzed with velocity vector. To assess the generality of our approach, a series of printing experiments were conducted with the metal droplet on the glass and copper substrates which were connected on a LT. Compared to the deposition on a static substrate, such the ultrasonic vibration had 31% and 2.1% improvement in the spreading diameters for the non-wetting system and wetting system and the corresponding contact angles were decreased by 23% and 11.2%. However, for the wetting system, the support of the ultrasonic vibration substrate at higher voltage and the suspension of the droplet could decrease the spreading diameter and increase the contact angle. Furthermore, the adhesion tangential force between the solidified droplet and substrate was measured, with 3.85 and 5.59 times increase in adhesion tangential force for the non-wetting system and wetting system. To concluded, the feasibility and effectiveness of the proposed UAMDD were verified by simulations and experiments. We expect that this technique can be extended to the droplet-based 3D printing of other metallic materials.

In the future, using the UAMDD method in metal droplet pileup should be further studied as the first results are promising. The effect of ultrasonic vibration is currently being investigated to increase the wettability and bonding strength for impacting droplets on non-wetting and wetting substrates. Furthermore, the results need to be verified for the two stacked droplets. In addition, the height uniformity of the droplet bumps controlled by ultrasonic vibration would be investigated to achieve uniform morphology and big bonding strength for the stacked droplets.

CRediT authorship contribution statement

Yuming Feng: Conceptualization, Software, Formal analysis, Validation, Investigation, Writing – original draft, Visualization. **Junkao Liu:** Writing – review & editing, Supervision. **Hengyu Li:** Software, Data curation, Validation, Writing – review & editing. **Jie Deng:** Software, Writing – review & editing, Funding acquisition. **Yingxiang Liu:** Methodology, Writing – review & editing, Supervision, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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